Lecture 25: General Property Equations from the Math of Thermo

CBE 20260 / Thermodynamics I

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1 Introduction: Calculating Thermodynamic Properties Using the Math of Thermo

In the last lecture, we derived the Maxwell relations and showed how to derive general expressions for the change in entropy with changes in (T, P) and (T, V). In this class, we will develop general expressions for the change in enthalpy and internal energy, review some basic "math tricks", introduce the concepts of volume expansivity and isothermal compressibility, and show a rigorous expression for $C_P - C_V$, valid for all substances.

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2 Calculating Enthalpy and Internal Energy

Now let's find the general equation for Enthalpy using T and P:

$$dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP$$

We already know the first term is C_p. For the second term, we use the fundamental equation $dH = TdS + VdP$. Dividing by dP at constant T:

$$\left(\frac{\partial H}{\partial P}\right)_T = T\left(\frac{\partial S}{\partial P}\right)_T + V$$

Substitute the Maxwell relation $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$:

$$\left(\frac{\partial H}{\partial P}\right)_T = V - T\left(\frac{\partial V}{\partial T}\right)_P$$

So the exact, general equation for enthalpy is:

$$dH = C_p dT + \left[V - T\left(\frac{\partial V}{\partial T}\right)_P\right] dP \tag{1}$$

2.1 Sanity Check: The Ideal Gas

Does this work for an ideal gas? Let's check. For an ideal gas, $V = \frac{RT}{P}$. The derivative is: $\left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P}$. Substitute this into the bracketed term in our dH equation:

$$\left[V - T\left(\frac{R}{P}\right)\right] = V - \frac{RT}{P} = V - V = 0!$$

The entire pressure dependence vanishes, leaving $dH = C_p dT$, exactly as we learned earlier in the semester! For a real gas, that bracketed term is *not* zero, because atomic interactions and excluded volume matter.

Similarly, we can derive the general equation for Internal Energy (U) as a function of T and V:

$$dU = C_v dT + \left[T\left(\frac{\partial P}{\partial T}\right)_V - P\right] dV \tag{2}$$

(*You should verify for yourself that the bracketed term goes to zero for an ideal gas!*)

We can use Eq. 1 to compute the Joule-Thomson coefficient for a real gas. Setting Eq. 1 equal to zero (a Joule-Thomson expansion is a constant entropy process) yields

$$C_p dT = \left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right] dP$$

$$\mu_{JT} \equiv \left(\frac{\partial T}{\partial P}\right)_H = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right]$$

You can verify that for an ideal gas, $\mu_{JT} = 0$

3 Mathematical Tools: The Cyclic Rule

To evaluate these partial derivatives for liquids and solids, we need a few more mathematical tools. A highly useful identity from multivariable calculus for any implicit function $F(x, y, z) = 0$ is the **Triple Product Rule** (or Cyclic Rule):

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1 \tag{3}$$

Let's apply this to our $P - V - T$ surface:

$$\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial V}{\partial P}\right)_T = -1$$

Rearranging this gives us a powerful way to evaluate the change in pressure with temperature at constant volume, which is notoriously difficult to measure experimentally:

$$\left(\frac{\partial P}{\partial T}\right)_V = -\frac{\left(\frac{\partial V}{\partial T}\right)_P}{\left(\frac{\partial V}{\partial P}\right)_T}$$

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4 Volumetric Properties: β and κ

For liquids and solids, engineers often tabulate volumetric behavior using two coefficients:

1. **Volume Expansivity (β):** How much a material expands when heated at constant pressure.

$$\beta \equiv \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P \tag{4}$$

2. **Isothermal Compressibility (κ):** How much a material shrinks when squeezed at constant temperature. (The negative sign ensures κ is a positive number).

$$\kappa \equiv -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T \tag{5}$$

If we substitute these definitions into the cyclic rule relation we just derived, we get:

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{\beta V}{\kappa V} = \frac{\beta}{\kappa}$$

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5 The Rigorous $C_p - C_v$ Relationship

Earlier in the semester, we stated that for an ideal gas, $C_p - C_v = R$. But what is the relationship for a real fluid?

Let's start by equating our two equations for entropy:

$$dS = \frac{C_p}{T}dT - \left(\frac{\partial V}{\partial T}\right)_P dP = \frac{C_v}{T}dT + \left(\frac{\partial P}{\partial T}\right)_V dV$$

Divide the entire equation by dT while holding volume constant ($dV = 0$):

$$\frac{C_p}{T} - \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial P}{\partial T}\right)_V = \frac{C_v}{T} + 0$$

Rearrange to solve for $C_p - C_v$:

$$C_p - C_v = T\left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial P}{\partial T}\right)_V$$

Substitute the cyclic rule substitution $\left(\frac{\partial P}{\partial T}\right)_V = -\frac{(\partial V/\partial T)_P}{(\partial V/\partial P)_T}$:

$$C_p - C_v = -T\frac{\left[\left(\frac{\partial V}{\partial T}\right)_P\right]^2}{\left(\frac{\partial V}{\partial P}\right)_T}$$

Finally, substitute our definitions of β and κ:

$$C_p - C_v = \frac{TV\beta^2}{\kappa} \tag{6}$$

Why is this profound?

1. Because T, V, and κ are always positive, and β^2 must be positive, C_p **is always greater than or equal to** C_v.
2. As $T \to 0$ (absolute zero), $C_p \to C_v$.
3. For liquids and solids, water being a notable exception at 4°C, β is very small, meaning $C_p \approx C_v$. This mathematically proves why we often just use a single "C" for liquids and solids!

You should verify that Eq. 6 reduces to $C_p - C_v = R$ for an ideal gas.